

BAULKHAM HILLS HIGH SCHOOL

YEAR 11

HSC ASSESSMENT TASK

December 2009

**MATHEMATICS
EXTENSION 1**

**Time allowed - 50 minutes
Plus 5 minutes reading time**

DIRECTIONS TO CANDIDATES

- Attempt ALL questions.
- Allocated marks are indicated for each question.
- All necessary working should be shown.
- Write your teacher's name and your name on the cover sheet provided.
- At the end of the exam, staple your answers in order behind the cover sheet provided.

Question 1

Find the Cartesian equation for the equation in parametric form $x = 2\sin\theta$, $y = 3\cos\theta$

2

Question 2

Find all real values of a for which $P(x) = ax^3 - 8x^2 - 9$ is divisible by $(x - a)$

2

Question 3

If α, β , and γ are the roots of $x^3 - 3x + 1 = 0$.

Find:

i) $\alpha + \beta + \gamma$

1

ii) $\alpha\beta\gamma$

1

iii) $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma}$

2

Question 4

The variable point $(3t, 2t^2)$ lies on the parabola. Find the cartesian equation for this parabola.

2

Question 5

For $P(x) = x^3 - 3x^2 - 3x + 10$

i) Show that $x = 2$ is a root of $P(x)$

1

ii) What is the product of the other two roots?

2

Question 6

i) Sketch the parabola whose parametric equations are

1

$$x = t, y = \frac{1}{2}t^2$$

ii) Clearly mark and label the directrix and the focal point

2

iii) Mark the point P and Q which correspond to $t = 1$ and $t = -2$

2

iv) Find the equation of the tangent at P and hence state the equation of the tangent at Q

3

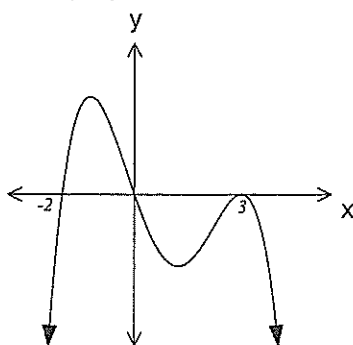
v) Show that the tangents intersect at $R(-\frac{1}{2}, -1)$

2

Question 7

Write down a possible equation of the polynomial with the following graph.

2

**Question 8**

- i) Express $F(x) = x^3 + 3x^2 - 10x - 24$ as a product of three linear factors
- ii) Sketch the curve
- iii) Hence solve, $x^3 + 3x^2 - 10x - 24 > 0$

3

1

1

Question 9

- A monic polynomial $P(x)$ of degree 4 has zeros at 2 and -2
- $P(0) = 4$
- and $P(1) = -3$

5

Find $P(x)$ and hence solve $P(x) = 0$ for all real roots.

Question 10

A chord PQ of a parabola $x^2 = 4ay$ is such that it is always parallel to the line $y = x$, where P is $(2ap, ap^2)$ and Q is $(2aq, aq^2)$

- i) Find the gradient of PQ and hence show that $p + q = 2$
- ii) Show that the equation of the normal at P is $x + py = 2ap + ap^3$
- iii) A normal is also drawn through Q. Find the point of intersection, R, of these two normals.
- iv) Find the locus of R.

2

2

3

2

Quest. 1. (2 marks)

$$\frac{x}{2} = \sin \theta \quad \frac{y}{3} = \cos \theta$$

since

$$\sin^2 \theta + \cos^2 \theta = 1 \quad (1)$$

$$\therefore \left(\frac{x}{2}\right)^2 + \left(\frac{y}{3}\right)^2 = 1$$

$$\left. \begin{aligned} \frac{x^2}{4} + \frac{y^2}{9} &= 1 \end{aligned} \right\} (1)$$

Quest 2. (2 marks)

$$P(x) = ax^3 - 8x^2 - 9$$

$$P(a) = a^4 - 8a^2 - 9 = 0$$

$$(a^2 - 9)(a^2 + 1) = 0 \quad (1)$$

$$a^2 = 9 \quad a^2 = -1$$

or each answer

$$a = \pm 3 \quad (1) \Rightarrow \text{no soln}$$

$$\therefore \underline{a = \pm 3}$$

Quest 3. (4 marks)

$$x^3 - 3x + 1 = 0$$

$$i) \alpha + \beta + \gamma = -\frac{b}{a} = 0 \quad (1)$$

$$ii) \alpha\beta\gamma = -\frac{d}{a} = -1 \quad (1)$$

$$iii) \alpha\beta + \alpha\gamma + \beta\gamma = \frac{c}{a} = -3 \quad (1)$$

$$\therefore \frac{\alpha\beta + \alpha\gamma + \beta\gamma}{\alpha\beta\gamma} = \frac{-3}{-1}$$

$$= \underline{+3} \quad (1)$$

Quest 4. (2 marks)

$$\text{let } x = 3t, \quad y = 2t^2 \quad (1)$$

$$\frac{x}{3} = t \quad y = 2 \times \left(\frac{x}{3}\right)^2$$

$$\text{sub in } \rightarrow y = \frac{2x^2}{9} \quad (1)$$

$$\text{or } x^2 = \frac{9}{2}y$$

Quest 5. (3 marks)

$$P(x) = x^3 - 3x^2 - 3x + 10$$

$$i) P(2) = (2)^3 - 3(2)^2 - 3 \times 2 + 10 \quad (1)$$

$$= 0$$

$\therefore x = 2$ is a root of $P(x)$

ii) product of roots

$$\alpha\beta\gamma = -\frac{d}{a}$$

$$2\beta\gamma = -10$$

$$\underline{\beta\gamma = -5} \quad (1)$$

(1) meth

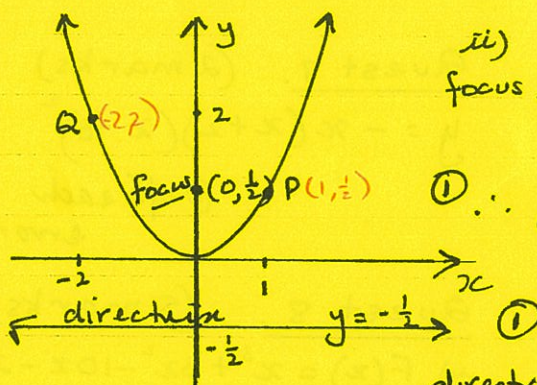
Quest 6. (10 marks)

$$i) x = t, \quad y = \frac{1}{2}t^2$$

$$y = \frac{1}{2}x^2$$

$$\text{or } x^2 = 2y$$

sketch (1)



$$ii) \text{ focus: } 4a = 2$$

$$a = \frac{1}{2}$$

$$\therefore (0, \frac{1}{2}) \quad (1)$$

$$\text{directrix: } y = -a$$

$$y = -\frac{1}{2} \quad (1)$$

$$iii) t = 1 \quad P(2ap, ap^2) = (1, \frac{1}{2}) \quad (1)$$

$$t = -2 \quad Q(2aq, aq^2) = (-2, 2) \quad (1)$$

iv) tangent at P. $(1, \frac{1}{2})$

$$y = \frac{1}{2}x^2$$

$$\frac{dy}{dx} = x \quad (1)$$

$$\text{at } x = 1 \quad m = 1$$

$$\therefore y - \frac{1}{2} = 1(x - 1)$$

$$y = x - \frac{1}{2} \quad (1)$$

at Q (-2, 2)

$$m = -2$$

$$\therefore y - 2 = -2(x + 2)$$

$$y = -2x - 2 \quad (1)$$

v.) $y = x - \frac{1}{2} \quad \dots (i)$

$$y = -2x - 2 \quad \dots (ii)$$

equating (1) & (2).

$$x - \frac{1}{2} = -2x - 2$$

$$3x = -\frac{3}{2} \quad (1)$$

$$x = -\frac{1}{2}$$

sub into (1)

$$y = -\frac{1}{2} - \frac{1}{2}$$

$$= -1 \quad (1)$$

$\therefore R(-\frac{1}{2}, -1)$

Quest 7. (2 marks)

$$y = -x(x+2)(x-3)^2$$

(1 off each error.)

Quest 8. (5 marks)

i) $F(x) = x^3 + 3x^2 - 10x - 24$

try 1, -1, 2, -2 etc.

$$F(3) = 27 + 27 - 30 - 24$$

$$= 0 \quad (1)$$

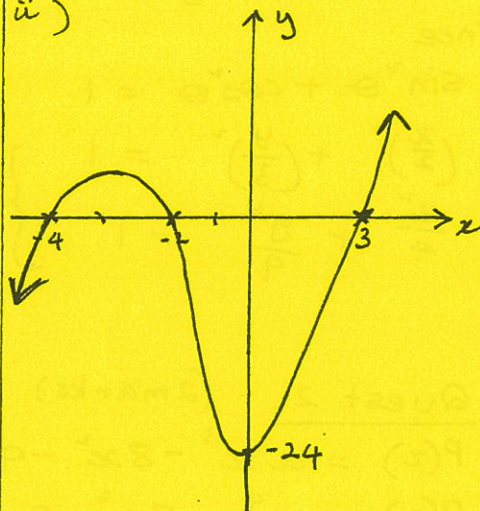
$(x-3)$ is a factor

$$(x-3) \overline{) \begin{array}{r} x^3 + 3x^2 - 10x - 24 \\ x^3 - 3x^2 \\ \hline 6x^2 - 10x \\ 6x^2 - 18x \\ \hline 8x - 24 \\ 8x - 24 \\ \hline 0 \end{array}} \quad (1)$$

$$\therefore F(x) = (x-3)(x^2 + 6x + 8)$$

$$= (x-3)(x+4)(x+2) \quad (1)$$

ii)



iii) $-4 < x < -2, x > 3. \quad (1)$

Quest 9. (5 marks)

$$P(x) = (x-2)(x+2)(x^2 + bx + c) \quad (1)$$

$$P(0) = (-2)(2)(c) = 4$$

$$-4c = 4$$

$$c = -1 \quad (1)$$

$$\therefore P(x) = (x-2)(x+2)(x^2 + bx - 1)$$

$$P(1) = (-1)(3)(1 + b - 1) = -3$$

$$-3b = -3$$

$$b = 1 \quad (1)$$

$$\therefore P(x) = (x-2)(x+2)(x^2 + x - 1)$$

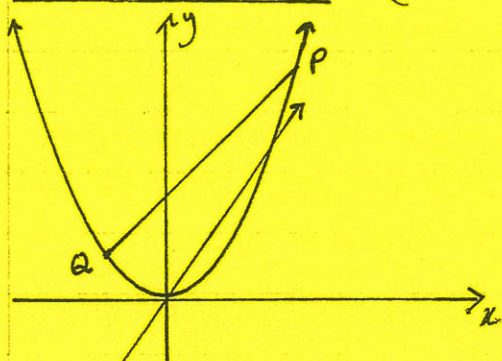
Solve $x^2 + x - 1 = 0$

$$x = \frac{-1 \pm \sqrt{1+4}}{2}$$

$$= \frac{-1 \pm \sqrt{5}}{2} \quad (1)$$

$\therefore$ Real roots of $P(x) = -2, 2, \frac{-1 \pm \sqrt{5}}{2} \quad (1)$

Question 10. (9 marks)



$P(2ap, ap^2)$ $Q(2aq, aq^2)$

$$\begin{aligned} \text{i) } m_{PQ} &= \frac{y_2 - y_1}{x_2 - x_1} \\ &= \frac{aq^2 - ap^2}{2aq - 2ap} \\ &= \frac{a(q-p)(q+p)}{2a(q-p)} \\ &= \frac{p+q}{2} \end{aligned} \quad (1)$$

now for $y = x$
 $m_1 = 1$

$$\begin{aligned} \therefore m_{PQ} &= m_1 \\ \frac{p+q}{2} &= 1 \\ p+q &= 2 \end{aligned} \quad (1)$$

ii) $x^2 = 4ay$

$$y = \frac{x^2}{4a}$$

$$\frac{dy}{dx} = \frac{x}{2a}$$

at $x = 2ap$ $m_1 = \frac{2ap}{2a} = p$

for normal $m_2 = -\frac{1}{m_1} = -\frac{1}{p}$ (1)

$$y - ap^2 = -\frac{1}{p}(x - 2ap)$$

$$py - ap^3 = -x + 2ap$$

$$x + py = 2ap + ap^3 \quad (1)$$

iii) at Q.

normal: $x + py = 2aq + aq^3$ (1)

R: $x = -py + 2ap + ap^3$ -- (i)

$x = -qy + 2aq + aq^3$ -- (ii)

subtract + substit.

$$-py + 2ap + ap^3 = -qy + 2aq + aq^3$$

$$qy - py = 2aq - 2ap + aq^3 - ap^3$$

$$y(q-p) = 2a(q-p) + a(q-p)(q^2 + pq + p^2)$$

① for working $y = 2a + a(p^2 + pq + q^2)$
 $= a(p^2 + pq + q^2 + 2)$

sub in for x . -- (1)

$$\begin{aligned} x &= -ap(p^2 + pq + q^2 + 2) + 2ap + ap^3 \\ &= -ap^3 - ap^2q - apq^2 - 2ap + 2ap + ap^3 \\ &= -apq(p+q) \end{aligned}$$

(1)

$\therefore R: (-apq(p+q), a(p^2 + pq + q^2 + 2))$ *
but $p+q = 2$ and $a((p+q)^2 - 2pq + pq + 2)$

either form OK here. *

$\therefore (-2apq, a(6 - pq))$ *

Locus.

iv) $x = -2apq$

$y = a(6 - pq)$

$-\frac{x}{2a} = pq$

sub in

① working.

$$y = a(6 + \frac{x}{2a})$$

$$2y = 12a + x$$

$$x - 2y + 12a = 0$$

$\therefore$ a straight line. (1)